LEVEL UP ACADEMY

Class 10 | CBSE | Science

ANSWER KEY & MARKING SCHEME

Pre-Board Mock 1 (CBSE pattern)

Chapter / Topic: Full syllabus (Chapters 1–13)

For teacher use only. Do not distribute to students.

SECTION A

Q1. Which of the following is a combination reaction? **[1]**

Answer: (a) $CaO + H_2O \rightarrow Ca(OH)_2$

Q2. The pH of a solution is 13. The solution is: **[1]**

Answer: (c) strongly basic

Q3. Which of these metals does NOT give hydrogen gas with dilute hydrochloric acid? **[1]**

Answer: (a) Copper (it is below hydrogen in the reactivity series)

Q4. The name of the compound CH_3CHO is: **[1]**

Answer: (c) ethanal (–CHO is the aldehyde group)

Q5. Which of these has ten molecules of water of crystallisation? **[1]**

Answer: (d) Washing soda, $Na_2CO_3 \cdot 10H_2O$

Q6. A few drops of phenolphthalein are added to sodium hydroxide solution. The colour of the solution becomes: **[1]**

Answer: (b) pink

Q7. Bile is poured into which part of the alimentary canal? **[1]**

Answer: (b) Small intestine

Q8. The correct path of urine in the human body is: **[1]**

Answer: (c) Kidney → ureter → urinary bladder → urethra

Q9. Which hormone prepares the body to deal with an emergency? **[1]**

Answer: (a) Adrenaline

Q10. Vegetative propagation by leaves is seen in: **[1]**

Answer: (d) Bryophyllum (buds in the notches along the leaf margin)

Q11. A cross between a heterozygous tall pea plant (Tt) and a short plant (tt) gives tall plants in the proportion: **[1]**

Answer: (b) 50%. Offspring are Tt and tt in a 1 : 1 ratio.

Q12. Which of the following is non-biodegradable? **[1]**

Answer: (d) DDT

Q13. A concave mirror forms a real image of the same size as the object. The object is placed: **[1]**

Answer: (b) at C

Q14. A wire of resistance 10 Ω is stretched uniformly to double its length. Its new resistance is: **[1]**

Answer: (d) 40 Ω. The length doubles and the area halves, so R becomes 4 times.

Q15. Which colour of white light is deviated the most by a glass prism? **[1]**

Answer: (c) Violet

Q16. The magnetic field lines inside a long, current-carrying solenoid are: **[1]**

Answer: (a) parallel straight lines (a uniform field)

Q17. **Assertion (A):** Rancidity of food can be slowed down by keeping it in airtight containers. **Reason (R):** Airtight containers reduce contact with oxygen,… **[1]**

Answer: (a)

Q18. **Assertion (A):** Pepsin acts on proteins in the stomach. **Reason (R):** The gastric glands secrete hydrochloric acid, which makes the medium in the… **[1]**

Answer: (a)

Q19. **Assertion (A):** A convex lens can form both real and virtual images. **Reason (R):** A convex lens always forms the image on the side opposite to the… **[1]**

Answer: (c). A is true, but R is false: when the object is between F_1 and O, the virtual image forms on the same side as the object.

Q20. **Assertion (A):** The concentration of pesticides is highest in organisms at the top of a food chain. **Reason (R):** Pesticides are non-biodegradable and… **[1]**

Answer: (a). This is biological magnification.

SECTION B

Q21. What happens when silver chloride is kept in sunlight? Write the balanced equation and name the type of reaction. **[2]**

Answer: White silver chloride turns grey. $2AgCl(s) \rightarrow 2Ag(s) + Cl_2(g)$ in sunlight. This is a decomposition reaction caused by light (photolytic decomposition), used in black-and-white photography.

- Observation + type: 1
- Balanced equation: 1

Q22. In the electrolysis of water, name the gases collected at the cathode and the anode. Why is the volume of one gas double that of the other? **[2]**

Answer: Hydrogen at the cathode and oxygen at the anode. Water (H_2O) has two hydrogen atoms for every oxygen atom: $2H_2O \rightarrow 2H_2 + O_2$, so the volume of hydrogen is double.

- Gases at electrodes: 1
- Reason with equation: 1

Q23. Give the function of: (a) villi in the small intestine, (b) salivary amylase. **[2]**

Answer: (a) Villi increase the surface area for absorbing digested food, and are richly supplied with blood vessels. (b) Salivary amylase breaks down starch into simple sugar.

- Each function: 1

Q24. Why do gametes have only half the number of chromosomes found in body cells? **[2]**

Answer: When two gametes fuse, the zygote gets the full set of chromosomes, so the chromosome number of the species stays the same in every generation. Otherwise the number would double each generation.

- Fusion restores full number: 1
- Keeps number constant across generations: 1

OR, answer: Every month the uterus prepares a thick, spongy lining to receive a fertilised egg. If the egg is not fertilised, the lining is not needed. It slowly breaks down and comes out through the vagina as blood and mucus. This happens roughly every month and lasts about 2–8 days.

- Uterine lining prepared: 1
- Breaks down when no fertilisation: 1

Q25. A 6 Ω and a 3 Ω resistor are connected in parallel, and this combination is connected in series with a 4 Ω resistor and a 6 V battery. Find the total… **[2]**

Answer: Parallel part: $1/R = 1/6 + 1/3 = 1/2$, so $R = 2\ \Omega$. Total $= 2 + 4 = 6\ \Omega$. $I = V/R = 6/6 = 1$ A.

- Total resistance 6 Ω: 1
- Current 1 A: 1

OR, answer: A concave lens of focal length $f = -80$ cm $= -0.8$ m. $P = 1/f = 1/(-0.8) = -1.25$ D.

- Concave lens: 1
- P = −1.25 D: 1

Q26. Why are some substances biodegradable and others non-biodegradable? Give one example of each. **[2]**

Answer: Biodegradable substances can be broken down by the enzymes of decomposers such as bacteria and fungi (e.g. fruit peels). Non-biodegradable substances cannot be broken down by these biological processes (e.g. plastic, DDT).

- Reason (enzymes of decomposers): 1
- Examples: ½ + ½

SECTION C

Q27. Describe the electrolytic refining of copper. **[3]**

Answer: Impure copper is made the anode and a thin strip of pure copper is made the cathode. The electrolyte is acidified copper sulphate solution. When current is passed, copper from the anode dissolves into the solution, and an equal amount of pure copper is deposited on the cathode. Soluble impurities go into the solution, and insoluble impurities settle at the bottom of the anode as anode mud.

- Anode/cathode/electrolyte: 1
- Pure copper deposits on cathode: 1
- Fate of impurities (anode mud): 1

Q28. A compass needle is placed near a straight wire carrying current. **[3]**

(i) The current in the wire produces a magnetic field around it, which acts on the compass needle. [1]
(ii) The deflection increases (the field becomes stronger). [1]
(iii) The deflection decreases (the field weakens with distance). [1]

Q29. An organic compound A with molecular formula $C_2H_4O_2$ turns blue litmus red. It reacts with sodium carbonate to give a gas that turns lime water milky…. **[3]**

Answer: A is ethanoic acid (acetic acid), CH_3COOH. $2CH_3COOH + Na_2CO_3 \rightarrow 2CH_3COONa + H_2O + CO_2$. The CO_2 turns lime water milky: $Ca(OH)_2 + CO_2 \rightarrow CaCO_3 + H_2O$

- Identify A: 1
- Each balanced equation: 1 + 1

Q30. Describe the path of blood through the human heart, showing how double circulation works. **[3]**

Answer: Deoxygenated blood from the body enters the right atrium through the vena cava, then goes to the right ventricle, which pumps it through the pulmonary artery to the lungs. Oxygenated blood returns through the pulmonary veins to the left atrium, then the left ventricle, which pumps it through the aorta to the body. Blood passes through the heart twice in each cycle, which is why it is called double circulation.

- Right side path: 1
- Left side path: 1
- Meaning of double circulation: 1

Q31. You accidentally touch a hot plate and pull your hand away at once. Describe the reflex arc involved. **[3]**

Answer: Heat receptors in the skin detect the stimulus. A sensory neuron carries the impulse to the spinal cord. A relay neuron in the spinal cord passes it to a motor neuron. The motor neuron carries the impulse to the arm muscles (effector), which contract and pull the hand away. The brain is informed later.

- Receptor + sensory neuron: 1
- Spinal cord / relay: 1
- Motor neuron + effector: 1

OR, answer: Nervous control uses electrical impulses through neurons, while hormonal control uses chemical messengers through the blood. Nervous control is very fast, while hormonal control is

slower. Nervous effects are short-lived, while hormonal effects last longer. Nervous impulses reach only the cells connected by nerves, while hormones reach all cells but act only on target cells.

- Any three: 1 each

Q32. A 2 cm tall object is placed 30 cm from a convex lens of focal length 20 cm. Find the position, size and nature of the image. **[3]**

Answer: u = −30 cm, f = +20 cm. 1/v = 1/f + 1/u = 1/20 − 1/30 = 1/60, so v = +60 cm (on the other side of the lens). m = v/u = 60/(−30) = −2, so h′ = −4 cm. The image is real, inverted and magnified (4 cm tall).

- v = +60 cm: 1
- Size 4 cm: 1
- Nature: 1

OR, answer: Real image, so m = −3. m = −v/u gives v = 3u = 3 × (−10) = −30 cm, so the image is 30 cm in front of the mirror. 1/f = 1/v + 1/u = −1/30 − 1/10 = −4/30, so f = −7.5 cm.

- m = −3: 1
- v = −30 cm: 1
- f = −7.5 cm: 1

Q33. An electric heater is rated 1000 W, 220 V. **[3]**

(i) $R = V^2/P = 220^2/1000 = 48.4\ \Omega$ [1]

(ii) $I = P/V = 1000/220 \approx 4.5$ A [1]

(iii) E = 1 kW × 2 h × 30 = 60 kWh (60 units) [1]

SECTION D

Q34. Answer the following: **[5]**

(i) Saturated hydrocarbons have only single bonds between carbon atoms (e.g. ethane, C_2H_6). Unsaturated hydrocarbons have a double or triple bond (e.g. ethene, C_2H_4). On burning, saturated hydrocarbons give a clean blue flame, while unsaturated ones give a yellow, sooty flame. [3]

- Difference with examples: 2
- Burning test: 1

(ii) $C_2H_4 + H_2 \rightarrow C_2H_6$, in the presence of a nickel (or palladium) catalyst. [2]

- Equation: 1
- Catalyst: 1

OR, answer: (a) Fix two nails on a cork in a beaker and connect them to a bulb and a battery through a switch. Pour in dilute HCl and switch on: the bulb glows. With glucose or alcohol solutions, the bulb does not glow, although they contain hydrogen. So acids produce H^+ ions in solution, which carry the current. (b) Distilled water has no ions. Rain water contains dissolved acids and salts from the air, which give ions that conduct electricity.

- (a) set-up 1, observations 1, conclusion 1
- (b) no ions in distilled water 1, ions in rain water 1

Q35. Answer the following: **[5]**

(i) Testes: produce sperm and the hormone testosterone. Scrotum: holds the testes outside the body at a lower temperature, needed for sperm formation. Vas deferens: carries sperm. Urethra: common passage for sperm and urine. Seminal vesicles and prostate gland: add secretions so the sperm are in a fluid, which also provides nutrition. [3]

- Any three parts with functions: 1 each

(ii) (Any two) Hair grows on the face (moustache and beard). The voice begins to crack and becomes deeper. Hair grows in the armpits and genital area. Sudden increase in height and weight. [2]

- Any two: 1 + 1

OR, answer: (a) Amoeba pushes out finger-like extensions (pseudopodia) that surround the food particle and form a food vacuole. Inside the food vacuole, complex substances are broken down into simpler ones, which diffuse into the cytoplasm. Undigested material moves to the surface and is thrown out. (b) Carbon dioxide from the air, through the stomata. Water (and minerals) from the soil, through the roots. Sunlight is absorbed by chlorophyll.

- (a) pseudopodia/vacuole 1, digestion 1, egestion 1

- (b) CO_2 source 1, water source 1

Q36. Answer the following: **[5]**

(i) In series the same current I flows through all the resistors, and the total voltage is the sum: $V = V_1 + V_2 + V_3$. By Ohm's law, $IR_s = IR_1 + IR_2 + IR_3$. Dividing by I: $R_s = R_1 + R_2 + R_3$. [3]

- Same current, voltages add: 1
- Ohm's law substitution: 1
- Final expression: 1

(ii) Parallel part: $1/R = 1/4 + 1/12 = 4/12$, so $R = 3\ \Omega$. Total = 3 + 3 = 6 Ω. $I = 12/6 = 2$ A. [2]

- Total resistance 6 Ω: 1
- I = 2 A: 1

OR, answer: (a) In parallel the voltage V across each resistor is the same, and the total current is the sum: $I = I_1 + I_2 + I_3$. So $V/R_p = V/R_1 + V/R_2 + V/R_3$, which gives $1/R_p = 1/R_1 + 1/R_2 + 1/R_3$. (b) Tungsten has a very high melting point (3380 °C), so it does not melt when it is hot enough to glow and give light.

- (a) same V, currents add 1, substitution 1, result 1
- (b) high melting point 1, glows without melting 1

SECTION E

Q37. Baking soda (sodium hydrogencarbonate) is commonly used in kitchens. When heated, it decomposes and releases carbon dioxide. If only baking soda is… **[4]**

(i) $NaHCO_3$ [1]

(ii) $2NaHCO_3 \rightarrow Na_2CO_3 + H_2O + CO_2$ [1]

(iii) On heating, baking soda leaves behind sodium carbonate, which tastes bitter. Tartaric acid reacts with the baking soda to release CO_2 without leaving sodium carbonate, so the cake rises and does not taste bitter. [2]

- Sodium carbonate causes bitterness: 1
- Acid neutralises / prevents it: 1

Q38. A student covered a healthy potted plant with a transparent polythene bag and tied it around the stem. After a few hours in sunlight, water droplets… **[4]**

(i) Transpiration [1]

(ii) The stomata in the leaves. [1]

(iii) It creates a suction (transpiration pull) that helps absorb water and minerals and move them up the xylem. It also cools the plant (temperature regulation). [2]

- Transpiration pull: 1
- Cooling: 1

Q39. Spherical mirrors are used widely in daily life. Dentists and people shaving use one type of mirror. Vehicles use another type as rear-view mirrors… **[4]**

(i) Concave mirror. When the face is within its focus, it gives an enlarged, erect image. [1]

(ii) It always gives an erect, diminished image and a much wider field of view. [1]

(iii) f = R/2 = +1.50 m, u = −5.00 m. $1/v = 1/f - 1/u = 1/1.5 + 1/5 = 13/15$, so $v \approx +1.15$ m. The image is 1.15 m behind the mirror (virtual, erect). [2]

- Correct signs and formula: 1
- v ≈ 1.15 m behind mirror: 1